

NAME

wait3, wait4 – wait for process to change state, BSD style

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <sys/wait.h>
```

```
pid_t wait3(int *_Nullable wstatus, int options,
             struct rusage *_Nullable rusage);
pid_t wait4(pid_t pid, int *_Nullable wstatus, int options,
             struct rusage *_Nullable rusage);
```

Feature Test Macro Requirements for glibc (see **feature_test_macros(7)**):

wait3():

Since glibc 2.26:

```
_DEFAULT_SOURCE
|| (_XOPEN_SOURCE >= 500 &&
    ! (_POSIX_C_SOURCE >= 200112L
        || _XOPEN_SOURCE >= 600))
```

From glibc 2.19 to glibc 2.25:

```
_DEFAULT_SOURCE || _XOPEN_SOURCE >= 500
```

glibc 2.19 and earlier:

```
_BSD_SOURCE || _XOPEN_SOURCE >= 500
```

wait4():

Since glibc 2.19:

```
_DEFAULT_SOURCE
```

glibc 2.19 and earlier:

```
_BSD_SOURCE
```

DESCRIPTION

These functions are nonstandard; in new programs, the use of **waitpid(2)** or **waitid(2)** is preferable.

The **wait3()** and **wait4()** system calls are similar to **waitpid(2)**, but additionally return resource usage information about the child in the structure pointed to by *rusage*.

Other than the use of the *rusage* argument, the following **wait3()** call:

```
wait3(wstatus, options, rusage);
```

is equivalent to:

```
waitpid(-1, wstatus, options);
```

Similarly, the following **wait4()** call:

```
wait4(pid, wstatus, options, rusage);
```

is equivalent to:

```
waitpid(pid, wstatus, options);
```

In other words, **wait3()** waits of any child, while **wait4()** can be used to select a specific child, or children, on which to wait. See **wait(2)** for further details.

If *rusage* is not NULL, the *struct rusage* to which it points will be filled with accounting information about the child. See **getrusage(2)** for details.

RETURN VALUE

As for **waitpid(2)**.

ERRORS

As for **waitpid(2)**.

STANDARDS

None.

HISTORY

4.3BSD.

SUSv1 included a specification of **wait3()**; SUSv2 included **wait3()**, but marked it LEGACY; SUSv3 removed it.

Including `<sys/time.h>` is not required these days, but increases portability. (Indeed, `<sys/resource.h>` defines the *rusage* structure with fields of type *struct timeval* defined in `<sys/time.h>`.)

C library/kernel differences

On Linux, **wait3()** is a library function implemented on top of the **wait4()** system call.

SEE ALSO

fork(2), **getrusage(2)**, **sigaction(2)**, **signal(2)**, **wait(2)**, **signal(7)**